Chapter 10

Managing a project

Today, the 'project' is the method of choice for organising most engineering work. As one engineer explained in an interview, 'We only do projects in this organisation, nothing else. Everything is a project.' There are, of course, other ways to organise engineering work. For example, operations and maintenance are typically organised on a continuing service basis. This kind of organisation is needed for running continuous process plants and may also be used in manufacturing. However, even within this kind of organisation, project organisation may still be used for specific time-limited activities, such as plant shutdowns for maintenance.

Your reputation, namely the extent to which people trust you to spend their money, depends on your ability to deliver on your promises. Every expert engineer has to be able to manage projects effectively and to deliver on promises. The challenge you will face is that delivery depends on the interpretations and subsequent actions of many people, working with incomplete information and large degrees of uncertainty in terms of actions and timing. This means that you have to reliably deliver predictable results in the face of ever-present uncertainty. How is that possible?

As explained earlier, there is a common misperception that engineering is a precise scientific activity with no room for uncertainty. It is easy for you to have gained this impression during your undergraduate studies, because in most engineering courses, uncertainty is only discussed in a very limited mathematical sense.

WORKING WITH UNCERTAINTY

Expert engineers deal with uncertainty all the time and have devised elaborate precautions to limit the consequences of that uncertainty as much as possible. It is partly their apparent success at doing so that leads people to think that engineering has become so predictable. At least in the industrialised world, for example, one never pauses to wonder whether the lights will come on when you flip the light switch.

We will discuss many ways to limit the consequences of uncertainty in this book. At this stage, it is important to appreciate some important ideas that can help correct common misconceptions for novice engineers.

One of the techniques for working with uncertainty is risk management.

Most students today now learn about risk management at some stage of their university course. It is worth remembering that the fundamental notion of a risk is associated with a particular unpredictable event that can affect outcomes, both

advantageously and disadvantageously. The magnitude of the risk depends on both the probability that the event will occur, and the consequences of its occurrence. The relative importance of a given risk, therefore, depends on a combination of probability and adverse consequences. Opportunities (uncertain events with favourable consequences) can be assessed in the same way. Different risks have different kinds of consequences. Some risks are primarily economic, such as an unexpected and significant rise in the price of a product from an engineering enterprise that dramatically improves profits. Others might be reputational risks, such as a high-profile employee being involved in court action related to allegations of misbehaviour. Still others might be technical risks: the possibility that a particular solution will not provide the performance that has been predicted. Risk management is a formal method in which engineers systematically examine all of the potential risks and opportunities, deciding on appropriate planning to maximise the opportunity to take advantage of 'upside' risks and to limit the undesirable consequences of 'downside' risks.

Now, let's look at some fundamental ideas behind the management of uncertainty in engineering.

Practice concept 56: Reliable performance requires a high level of predictability

Student engineers often think that . . .	Expert engineers know that . . .
50% is okay to pass and 70% is a distinction. Getting it 100% right is for nerds.	99.99% predictability is required (or more) for reliable performance.

This almost seems trivial. However, the grading system that governs your behaviour as a student means that you rarely get the chance to learn how to perform engineering technical work to an extremely high level of predictability and reliability. You need to remember that engineers work under similar time pressure as you did when you were a student. There is simply never enough time to do everything. How, then, does an expert engineer manage to achieve such a high level of reliability in technical work performed by people who are no more predictable in their professional working life than they were as students?

As unbelievable as that might sound, you can also learn how to achieve this.

Practice concept 57: Uncertainty in engineering mostly arises from unpredictable human behaviour

Novice engineers often think that . . .	Expert engineers know that . . .
Uncertainty is a probability, most often following a statistical normal distribution.	Uncertainty is mostly due to unpredictable differences in human interpretation, subsequent actions, and timing.

To the extent that uncertainty is part of an engineering curriculum, it is almost always considered to be a statistical phenomenon, often receiving very little attention in most subjects that make up the engineering course. It may arise in the form of 'noise', 'signal to noise ratio', or 'the Weibull distribution, which describes the distribution of component failures over time'. Uncertainty is only described in terms of inanimate causes. If people are considered at all, the consequences of human behavioural uncertainty can be minimised by rules, protective clothing, fences, gates, warning signs, regulations, and laws.

At least until the first half of the 20th century, most of the uncertainty faced by engineers was due to natural causes or limitations in our knowledge of engineering science. Today, much of that has changed. Engineering science now allows us to make extraordinarily accurate predictions . . . most of the time. Obviously, engineered structures can still be overwhelmed by natural disasters, such as storms, floods, volcanic eruptions, earthquakes, tsunamis, and wildfires. However, these events are very rare in statistical terms. Much more commonly, today at least, uncertainty and limitations on predictability arise from human behaviour and interpretation. The 2011 Japanese tsunami was predictable to natural scientists. However, people had decided that the risk was so small that it was not worth the expenditure to create sufficiently strong defences.

Humans are intrinsically unpredictable, though we all have mostly predictable aspects, as well. Engineers and all the other people who work in engineering enterprises are no more predictable than other people.[1]

Therefore, expert engineers have a large number of special techniques that take human characteristics into account to ensure that the results of engineering are reliably predictable.

Risk management practice helps to confirm the overwhelming influence of human behaviour in contributing uncertainty to contemporary engineering projects. A typical risk assessment for a medium-sized project might list 200 or more sources of uncertainty, of which 5–15 will normally originate from natural variability, and perhaps two or three will be completely unpredictable. For example, weather is a natural source of uncertainty for any outdoor engineering activity, but modern forecasting almost always enables bad weather to be predicted days in advance, long enough to avoid any risks. The other 190 or so uncertainties will involve human unpredictability, such as failure to perform maintenance, obtaining the wrong fuel or materials, not following appropriate procedures, tired operators who are more likely to make mistakes, running out of spare parts, collisions involving vessels, aircraft or vehicles controlled by people, and many others.

As we shall see below, expert engineers know that understanding how human behaviour contributes to uncertainty is a critical aspect of practice. Every day, engineers have to confront uncertainty, most of it originating from human behaviour. However, some of the best ways to reduce the consequences of uncertainty lie in making use of the same human diversity that leads to uncertainty in the first place.

How does human behaviour cause uncertainty, you may ask?

Remember that technical work is largely autonomous: it relies extensively on human interpretation, human cognition, and human action, as well as social interactions. It is not uncommon to find people in an engineering enterprise with 60 or more simultaneous issues to deal with: requests for assistance, coordination of other

people, decisions that have to be made, collecting information to help with decisions, e-mail correspondence, and administrative issues. What people do each day, the order in which they do it, when they choose to start, what they remember to do and what they forget to do, when things get interrupted by the actions of others, when people forget to return to actions that were interrupted: all these contribute to the unpredictability in engineering work performed every day. Beyond those common examples, there are human errors or mistakes. These can be unconscious slips made without any awareness, deliberate incorrect actions taken because of incorrect perception, deliberate incorrect actions taken because of interpretation differences, deliberate incorrect actions taken because of habit, failure to acknowledge incorrect actions and take corrective action in time, and incorrect actions taken with the intent to deceive and keep them hidden.[2]

Engineering techniques to limit the consequences of human uncertainty include education (to understand the purpose of the enterprise or project), skill development and training, organisational procedures and checklists, risk management, project planning, individual skill, experience and knowledge, such as how to prevent interruptions that lead to slips, practice to perfect skills, technical standards to guide action, monitoring, the use of appropriate tools, appropriate organisational culture and attitudes towards quality, and documentation.

Engineering enterprises are also vulnerable to unpredictable human behaviour from external causes, such as changes in the economic or political environment, changed financial circumstances affecting investors, colleagues, contractors, and suppliers, industrial action by trade unions, and the consequences of various actions that spill over and affect other people.

Practice concept 58: Human diversity can help reduce behavioural uncertainties

One of the most effective ways to limit the consequences of unpredictable human behaviour is to make use of the diversity in perception, interpretation, thinking, and action that is intrinsically human. Different people often see the same thing differently: that is why it is best for documents to be checked by someone other than the authors.

The mere expectation that work will be systematically checked by senior engineers can improve performance by raising expectations on the expected standard of work.

Novice engineers and students seem to be very reluctant to engage in this. In our research, we have observed how many experienced engineers, judging from their actions and interview responses to carefully designed questions, think that systematic checking is a non-productive chore for which, they say, they don't have enough time.[3] Without understanding how reducing uncertainty contributes to value perceptions, it may not be easy for engineers to understand how checking produces any real value. In practice, the earlier a mistake is detected, the cheaper it is to correct. Something that might take four hours in a design office might require hundreds of hours to fix if it remains undetected until construction is half-completed. One of the factors that most often contributes to unexpected project delays is discovering mistakes that should have been detected earlier.[4]

Novice engineers often think that . . .	Expert engineers know that . . .
Checking is just a chore that takes away valuable, productive time. The more you check, the more mistakes you will find.	As an example, systematic checking is essential and best done by another person, as they will see things in a different way. This helps expose alternative interpretations and reduces the chances of undetected mistakes, thereby increasing value by reducing uncertainty.

The same principle can be applied in risk management. Involving non-engineers and external people in the assessment of environmental and social risks, in particular, is now part of established practice.[5] Non-engineers and outsiders can often spot mistakes and omissions made by engineers much easier than engineers can and can point out some of the unexpected social implications of what otherwise might seem like optimal technical solutions. They are not necessarily any smarter than engineers: they just think in different ways and have different ways of perceiving the world. In other words, they see things differently. They can think of obvious things that can go wrong that engineers easily overlook.

Later, we will look again at review processes and checking in engineering practice.

Practice concept 59: Project management enables predictable results in the presence of uncertainty

Novice engineers often think that . . .	Expert engineers know that . . .
Project management consists of planning, preparing a Gantt chart, and is largely an administrative non-technical function, something that anyone who can't handle 'technical stuff' is able to do.	Project management means leading and guiding people to faithfully translate technical ideas into an engineering reality in line with project objectives.

Project management means coordinating an enterprise to reliably deliver predicted results, on time, safely, and within the planned resource budget. As an expert engineer, if you can reliably deliver predicted results in the face of uncertainty, then people will trust you with a lot of their money. It is not easy to do this without knowing how. In practice, as we shall see, it is often impossible to achieve every project requirement; instead, a project manager may have to choose between multiple conflicting requirements.

Many engineers who prefer to focus on building their technical capabilities dismiss project management as something that anyone can do, particularly anyone who is not a technical expert. 'They're just project managers; they can't hack the technical stuff, so they get other people to do it for them.'

Sooner or later, however, every expert engineer who focuses on technical subtlety or complexity comes up against the challenge of bringing ideas to reality at a scale that is beyond the capacity of one or two people, sometimes on a gigantic scale with a budget in the billions of dollars.

That's when project management becomes vitally important: it is the main process by which engineers deliver results of lasting value, the critical delivery thread of engineering practice, which was the large bottom half of Figure 3.7.

You will find project managers working in every organisation at every level. As the discipline has evolved, some people have emerged as project management specialists: project managers who find themselves able to manage any kind of project. Many engineers have encountered frustrations in working with professional project managers in charge of the projects in which they find themselves. This has arisen partly because engineers, as a professional group, have largely forgotten that delivering results is all about coordinating large numbers of people. This doesn't happen simply by writing documents and issuing them. It takes persuasion and gentle influence, also known as leadership, to guide people so that they all take part in a well-coordinated performance.

Some may argue that it is better for people with good relationship skills to take on project management responsibilities and leave engineers to do what they are better at, such as technical design, calculations, and modelling. However, there are three weaknesses with this argument. First, engineers need to develop 'influencing skills' in order for their ideas to be accepted and implemented faithfully, as we have already seen and discussed. Second, as explained in Chapter 4, anyone with the proper determination can become an expert: even engineers can become expert influencers and project managers, provided they are willing to engage in deliberate practice and seek the help of experts who can evaluate their performance. Third, technical understanding is helpful, if not essential, in delivering complex technical engineering work on time and budget.[6]

Almost all formal courses on project management focus primarily (often exclusively) on the production of documents, plans, and schedules. Unquestionably, that is an important element. Effective project management is a formal process and documentation is a part of it. However, that is probably the easiest part of project management. The more difficult part is not always included: dealing effectively with uncertainty through a formal risk management process. A good risk-management plan will provide a means to deal with almost all foreseeable, but unpredictable, events that can have a significant effect on the project outcome.

The truly difficult part of project management is making sure that the work runs according to the plans: it frequently does not, so the plans have to be modified to ensure that the objectives are still achieved. Therefore, an effective project manager knows how to draw up plans that can be changed, and how to monitor the progress of work and detect unwanted delays early enough to do something about them. This can be difficult, as many people do not want to even admit to themselves that they are running late. Understanding the different ways that people behave is essential for effective monitoring.

Finally, a good project manager knows how to set up systems to ensure that everything is checked, which ensures that when the project is completed, there is nothing that has been forgotten.

Obviously, financial controls are just as important as controlling the progress of technical work. As we shall see later in this chapter, monitoring finance is a critical part of monitoring the progress of the project as a whole. There are many technical and other techniques to learn: they will all come in a later chapter. Understanding why people behave the way they do helps you make the most of these techniques.

There have been many debates on the extent to which project managers need to have an understanding of technical issues. However, that argument is irrelevant for this book. Expert engineers need to be able to understand both technical issues and project management issues. Understanding how they overlap and how you can take advantage of that dual knowledge is something that you will develop with time and patience.

An engineering project starts with negotiations to clarify the project objectives and the client's requirements, particularly to understand the client's tolerance for uncertainty and financial risk. The technical aspects follow the typical sequence shown in the stack of blocks up the middle of Figure 3.10. These blocks describe activity by people which, as we have seen, involve countless unpredictable elements, people who are often tired, bored, anxious to get home, or simply distracted by other priorities.

Project management is a systematic combined performance that enables engineers to obtain predictable results in the presence of all this human and natural uncertainty. Project management happens in the big cylindrical tube surrounding the stack of blocks in the centre of the diagram: part of the tube has been cut away so that we can see the stack. Informal coordination is portrayed on the left side of the tube, while formal project management is portrayed on the right side (along with many other engineering management systems, of which project management is just one).

In essence, project management provides the guidance that keeps the technical engineering work on track and properly coordinated: the thin threads joining the tube to the stack represent this guidance, influence, collaboration, and coordination.

Figure 3.10 separates the core technical activities from the coordination and management that is taking place outside: this is purely to help understand the notion of a combined performance. Part of the work involves coordinating the performance. In reality, however, it is often the case that the same people perform both the technical and project management tasks.

The previous chapter described technical coordination: an informal process used by engineers all the time, usually in one-to-one collaboration relationships with other people. Technical coordination is usually undocumented, except for occasional diary entries and e-mail correspondence, in certain cases.

Project management, on the other hand, is a more formally documented process invoked when the scale of activity in cost, complexity, space, and time goes beyond what a single person can effectively coordinate in an informal manner. Project management heavily relies on written documents to supplement human memory.

Practice concept 60: Written knowledge needs to be learnt

While project management relies on documents, it is essential to understand how documents are used. Many of these documents provide the scripts for the overall combined performance. Most people in engineering enterprises, including most engineers, are reluctant readers, at best. Providing information in documents, no matter how well

they are written, is often as good as keeping the information secret. The information can only influence actions when it is in people's heads, and that requires learning.

Therefore, in the context of project management, documents mostly serve only as a record of discussions, which are often discovery performances by people (see Chapter 8) to build up knowledge and understanding about the project in their heads. Documents can extend human capability by storing information that is difficult to remember, but this is only useful if a person knows where the information can be found when it is needed. Documents also provide evidence to other people about the issues considered during the planning process. This can be important for seeking approval from government regulatory agencies later in the planning phase.

There are many different ways to approach project management: there are countless styles of documentation, planning techniques, and software systems for storing information. Most engineering enterprises have their own preferred ways of doing it. I am not going to describe any of these preferred ways, as you will learn them in due course. Instead, I'm going to take you through a long sequence of questions to make you think about how we can achieve a predictable result from a very large number of individual human performances, each of which has a certain degree of unpredictability. All of these questions are relevant when managing an engineering project.

You have to do most of the work: you will need to find answers for these questions and learn how to implement the answers within the project management system used by your enterprise.

Most likely, you are probably reading this chapter because you are facing some kind of project management responsibility. Your job is to seek the answers to these questions that are pertinent to your situation, here and now, by discussing them with your colleagues. Writing down these responses will provide a record of those discussions, evidence that they took place, and proof that your colleagues have worked with these ideas. The knowledge is therefore with them too. By the time you finish, you will have a workable plan, along with having learnt a lot about project management and how it fits into engineering practice.

Practice concept 61: Project management performances can be improved

Engineers around the world have an unenviable record in engineering project delivery, particularly in the energy, manufacturing, and chemical process industries that are increasingly critical for environmental sustainability. Performance is particularly troublesome in many developing countries, and larger projects are more likely to expose project delivery weaknesses

Ultimately, these failures are failures in engineering project management: a failure to deliver the promised results on time, on budget, and with predicted performance, safety, and reliability. It is possible that the many fundamental misconceptions related to engineering practice described in this book contribute to these failures. We have seen many instances of the misconceptions in this book directly contributing to project failures. In Chapter 11, we will closely examine the failure of a major engineering project to expose more of these issues.

While every guide to project management recognises the importance of communication, most restrict the understanding of communication to simple information transfer, extending from the project manager to the participants. Remember Practice Concept 40 in Chapter 7? If only everyone would simply read the plan and follow it!

Fortunately, by reading the last five chapters, you should now understand that effective collaboration between people relies on much more than that. It is not easy for information in written documents (or e-mails) to be learnt by people so that their actions are influenced appropriately. Building effective working relationships lies at the heart of collaboration, and building relationships requires much more attention to the idea that we are working with people, all of whom have different ideas and different interests. Influencing perceptions is crucial for influencing the actions of other people. In other words, we need a much richer understanding of communication.

Working with other people, especially working with lots of other people, can be a highly stressful and draining experience if you don't know how to do it properly. Like anything else, however, as you develop more expertise and you understand more, it can become one of the most enjoyable, rewarding, and simultaneously intellectually challenging aspects of engineering. Once you start to do it well, the improvement in your abilities will be impossible to miss.

Practice concept 62: The language of project management shapes behaviour

Project managers have evolved their own language and project management has become a discipline in its own right. There are some very interesting terms in contemporary project management literature.

The first is *resources*.[7] Project managers include other people, along with equipment, machinery, space, and consumable materials as 'resources'. For decades, 'resource allocation', 'resource levelling', or 'resource optimisation' have been a part of the vernacular of project management. That being said, recent research has started to draw attention to one of the drawbacks of referring to people as 'resources'.[8] People referred to in this way start to behave like inanimate resources. They don't move without being pushed. They resist efforts to push them and prefer to stay in one location. This resistance is silent: people don't complain and they don't talk about it, but they also don't respond.

That is why in this chapter, I do not refer to people as 'resources'. Instead I use the term 'human effort': work that has to be performed by people. When planning a project, we try and estimate the amount of time that will be needed, which is ostensibly the number of hours of human effort. However, in the last chapter, we saw how the number of hours of human effort is often completely unrelated to the nature of the task. People find effective ways to finish work quickly when they want to and other ways to make the work last much longer in different circumstances. The extent to which they have a positive relationship with the person coordinating and leading the work has a large influence on this factor, as described in Chapter 9.

Another interesting term is *subject matter expert*, a term used to refer to technical specialists such as engineers in project management literature. Subject matter experts are consulted and provide information, but seldom take any active role in decision-making. You should learn about project management so you don't get sidelined as a

subject matter expert. A 2004 survey of several thousand civil engineers in the USA revealed extensive dissatisfaction when engineers found themselves working under 'unqualified' project managers. The difficulty is that many engineers do not learn the appropriate skills to deliver practical results from their work and can therefore easily fall into this trap.

PLANNING, ORGANISING AND APPROVAL

Managing a project, like technical coordination, can be described as a three stage process:

1. planning and organisation,
2. monitoring progress, and
3. completing the project.

Most of the chapter is devoted to project planning, emphasising that a managing a project requires a detailed plan.

Practice concept 63: A project plan is a living document

The next part of the chapter presents a series of questions (italics headings). Several have sample answers (in italics) that will help you supply your own answers. The aim is to help you develop your project-planning skills by following a project in which you may already be involved. By seeking answers from your workplace colleagues in a series of discussions, which will likely be 'discovery learning performances', you can learn how to prepare your own project plan.

As you answer the questions, you are going to start to construct a project plan as the main practice exercise for this chapter.

However, even the best project managers don't get everything right the first time. Instead, every document that you create is a *living document* that evolves and is progressively elaborated during the course of the project as you, your colleagues, and often the client and other stakeholders get a clearer idea of the requirements. Some documents are still being elaborated even at the very end of a project.

A well-run project team will regularly review the project scoping document, specifications, and project plans throughout the project, every few days, weeks, or months, depending on the project duration, especially when the scope, specifications, or plans need to be changed.

Particularly when you are planning a project for the first time, be prepared to revisit all of these questions many times over. Each time you come back to these questions, you will find that you need to elaborate each of the documents that were created along the way. You will think of more details to add, just as in writing a chapter of a book, I think of more details to add to the text every time I look at it. However, engineering is pragmatic: there is always an endpoint to a project. There is always a time to move on and accept that we have done the best we can, but it is never complete.

At the same time, always look for ways to simplify the requirements without losing detail: keep documents as short as possible.

Defining the project scope and creating a plan: the project definition phase

What is a project?

This is one of the few questions with an answer! Be warned.

A project is a temporary endeavour undertaken to create a unique product, service, or result.[9] A project is temporary because every project has a definite beginning and a definite end. The end is reached when the project objectives have been achieved, it becomes clear that the project objectives will not or cannot be met, or the need for the project no longer exists and the project is terminated.

A project creates unique *deliverables*: products or physical assets, information, services, or results.

A project can create:

- A product (artefact) that has measurable, quantifiable attributes and can be either an end item in itself or a component of another item,
- The capability to perform a service, such as business functions supporting production or distribution, or, the performance of a defined service.
- A result such as information, outcomes, or documents. For example, a research project develops knowledge that can be used to determine whether or not a trend is present or if a new process will benefit society.

Uniqueness is an important characteristic of project deliverables. For example, thousands of office buildings have been developed, but each individual facility is

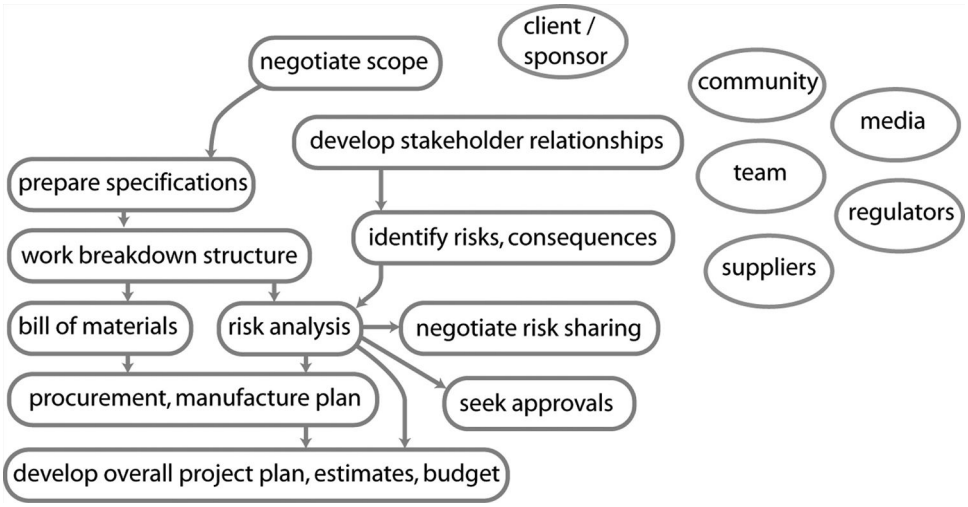


Figure 10.1 Project definition and planning phase. While there is a clear sequence, in reality, the process is at least partly iterative. As significant technical issues and uncertainties become clearer and as more investigation by the project team provides insight, it may be necessary to renegotiate the scope and risk sharing. It may even be necessary to revisit some of the planning once the project gets under way, as shown in Figure 10.3. Most of the major blocks in the diagram appear as subject headings in the text that follows.

unique – different owner, different design, different location, different contractors, and so on. The presence of repetitive elements does not change the fundamental uniqueness of the project work.

Do you have access to the planning documents for a previous, similar successful project?

Never start planning from scratch if you can adapt aspects of planning documents produced for earlier successful projects. Learn from others who have done that type of work before.

Negotiate and define the scope

This is the single most important job for a project manager: negotiating and reaching an agreement on defining the scope with the client and other stakeholders (see below). Some projects do not involve a paying client: they may be sponsored by someone with appropriate authority in your own firm, or may be a part of your education, in which case the sponsor is your lecturer. The sponsor instigates the project and can therefore be treated as equivalent to the client.

Mostly, especially in engineering projects, the sponsor or client has only a sketchy idea of the requirements. In the early phase of the project, therefore, your main role will be to find a way to meet the client's requirements and specify the necessary details.

Project scope is defined in one or more documents that provide a succinct definition of everything that is to be accomplished by a project.

While everything else can be progressively elaborated, be extremely cautious about elaborating anything in the project scope. This is called 'scope creep', a series of small extensions to the scope, each one being apparently insignificant. We will come to this issue again. Adding anything to the project scope almost invariably means increasing the cost and time to complete the project. Finalising the project scope definition right from the start is one of the most crucial skills of a project manager.

What is the project going to achieve? What are the project objectives?

Examples:

'Improved access to our factory to reduce the time and cost of unloading incoming parts, materials, and supplies, while also reducing the cost of loading finished products for delivery to customers.'
'Remove hazardous waste material from one or more designated locations and safely dispose of the material using appropriate methods.'
'Design, construct, and evaluate the performance of, and customer attitudes to, a novel prototype machine.'
'Evaluate the feasibility of implementing a new automated classification algorithm in a computer vision inspection machine.'

Answering this question, like many others in this chapter, requires a discovery learning performance. One or more discussions with the client or sponsor will help both parties learn about the objectives. Why is this needed for the client or sponsor who,

presumably, must have thought about this earlier? Often, they have not actually clearly stated the project objectives in terms that make sense in an engineering context. Most of the time, they will not be engineers and will therefore need your help to do this.

How will you know that you have achieved these objectives? What are the acceptance criteria?

Write a carefully worded statement to explain how you will be able to confirm that all the project objectives have been completely achieved. Whose opinion will be most important if there is any disagreement?

Once again, this may be part of the discovery performance with the client or sponsor.

What will be delivered to the client, or what task will be performed for the client (tasks and deliverables)?

Write a list of every item that needs to be delivered to the client, and when it is needed. Several items will probably have to be delivered well before the end of the project, especially documents, such as progress reports.

What are the important attributes of project performance (the project drivers)?

Schedule, quality, cost? Which of these three is the main *driver* for this project? The project plan and budget will be strongly influenced by the need to:

a) keep to a particular time schedule,
b) maintain certain quality standards, or
c) minimise the initial capital expense.

You need to understand which of these drivers is most important for the client. These are the three basic drivers; project performance attributes and the cost implications of each one will be explained in more detail in Chapter 11. Another driver is the degree to which the client has defined the project objectives. If the objectives are only vaguely defined and can't be tied down until later, it can be very difficult to forecast the actual budget without a large allowance to account for cost estimate increases once the requirements become clearer at a later date.

What are all the main activities that need to be completed to achieve the project outcomes?

Later, we will create a work breakdown structure (WBS) which is a complete list of work activities to be performed by everyone concerned with the project. At this stage, it is only necessary to list the major activities that will be part of this. Remember to include documents, procedures (work needed to secure relevant government and regulatory approvals), and document checking by the client.

What will the client pay for?

It is easy to assume that the client will pay for all activities, but this may not actually be the case. The client may have assumed that certain activities will be paid for by other stakeholders, such as government agencies, or even the contractor.

What are the main activities required for the project that the client is not paying for and who is going to pay for them?

This is crucial: a project manager must know who is paying for all the activities needed to complete the project.

Are there any main activities that no one is going to pay for?

If there are, find a way to remove them completely from the project scope.

Note that the project scope does not have to specify every single activity in the project. The project scope is always written with the client in mind: put yourself in the position of the client and ask yourself the same questions again. Review your answers and, if necessary, reword them as the client would write them.

What is the time schedule for the deliverables or services to be performed?

- When does the client expect the outcomes of the project to be delivered?
- Where are the *deliverables* to be provided?
- In what form are they to be provided?
- If they are documents, will they be in electronic or printed form?
- If electronic, which formats are to be provided?
- Who is responsible for the indefinite storage of a reference copy of each of the documents? (This is particularly important for electronic documents.)

What is the project budget?

As I will explain in Chapter 11, there is nothing simple about estimating project costs and setting an appropriate budget. Cost is affected by other aspects of the requirements: the need to keep to a particular schedule and the need to maintain certain quality standards. There is also the definition of project cost: is the client talking about the total life-cycle cost, including costs incurred long after the project is completed, such as maintenance and operating expenses? Or is the client simply referring to project cost, and wants to minimise the expense during the project itself, regardless of costs incurred later as a result of doing that?

Expect that project management activities, such as defining the project, developing the project plan, and monitoring the project after the work has started will cost 2–4% of the total project budget.

Also, take into account that it may not be possible to accurately forecast the cost of the project until most of the plan is in place, and at least part of the detailed engineering design has been completed. Sometimes, when a project is repeating something very similar that has been done before, accurate estimates of the total cost may be possible based entirely on size parameters of the objectives, such as the length of a freeway given the number of traffic lanes or the number of drawings that need to be completed for a structural design project.

Apart from financial resources, what is the client providing for the project?

- Are there people who work for the client who have to be involved directly in the project?
- Is the client going to provide land, space, materials, equipment, information, or software?
- What are the expectations for this?
- When are these resources going to be provided?

Is there a possibility that these resources may not be available when promised?

- If so, how will equivalent resources be provided for the project when needed?

What is the contractual arrangement between the project team and the client?

- If there is no existing contractual arrangement, how will one be prepared and put in place?

How much is the client going to pay and when?

- What proportion of the total cost is to be paid in advance of any work being completed and accepted as complete?
- What precautions can be taken to minimise the risk that work that has been paid for in advance cannot be completed by the contractor(s)?
- If the client is going to make payments before the project is completed, what conditions have to be satisfied before the client makes these payments?

Practice concept 64: Specifications

The scope is not complete without specifying details of the artefacts, information, or particular services that will be provided for the client. Often, the specifications are separate documents referred to by the scope document. The specifications may be written by the project team and negotiated with the client early in the project but gradually refined throughout the course of the project. Alternatively, they may be provided at the start of the project as non-negotiable documents.

Often, the specifications are part of a contract defining the project, binding the enterprise delivering the project – remembering that we are using the term 'enterprise' to include the client, end users, and the organisations that actually perform the work.

The specification usually does not define every detail; instead, it provides sufficient technical information so that the deliverables will be fit for the intended purpose(s). Like the scope, it is often up to the project team to write the specifications and then reach an agreement with the client or sponsor.

The specifications will also refer to drawings, or, more likely, digital representations of the object to be created, encompassing most, if not all, aspects of the design.[10] These representations of the complete object also provide the details needed to construct the *bill of materials* and *work breakdown structure*, both of which will be described later.

So, the next logical question is: how is a specification structured as a written document?

For this, we need to understand the purpose of a specification: it defines what is acceptable and how we can determine whether the particular deliverable is acceptable or not.

Remember Practice Exercise 9, when you attempted to write an outline specification for a beam? A real specification has to include a definition that will allow an engineer to decide if the artefact, information, or service meets the requirements.

Although the client does not need to understand every detail contained in the specification documents, the people who are responsible for delivering the artefact, material, service, or information will need to understand. As an engineer, one of your primary responsibilities is to decide the level of detail required for the specification document in order for the client's requirements to be met at the lowest possible cost. Chapter 11 includes a case study that shows how a minor error in the level of detail included in a specification resulted in additional costs that amounted to millions of dollars for a single item of instrumentation.

There are two basic types of specification: test and method.

Test specification

The first type of specification is known as *test specification*: we describe the testing and inspection that will confirm that an objective has been achieved (in other words, the result complies with the specification) or has not been achieved (because the result does not comply with the specification).

For example, we might be required to construct a beam. The specified acceptance test might consist of a defined loading on the beam and measurement of its deflected shape. Hence, the specification might define:

a) how the beam is to be positioned and supported,
b) the applied load(s) and how the loads are to be applied (e.g. a testing machine, weights, attached steel ropes, etc.),
c) allowable deflection limits in certain directions and rotational movements at given positions on the beam,
d) the length of time for which the load is applied,*
e) maximum and minimum temperature and relative humidity limits for testing,*
f) storage conditions for the beam prior to test,* and
g) the maximum allowable wind (if the test is done outside).

Items marked with an asterisk (*) are important for non-metallic structures, particularly fibre-reinforced composites, plastic materials, or natural materials, such as wood.

If possible, it is preferable to use standard tests so the results can be compared with previous tests. For example, the American Society for Testing and Materials (ASTM) is an international standards organisation that develops and publishes voluntary consensus technical standards that focus on testing methods; they publish a large catalogue of testing standards.

The weakness of a test specification is that the testing is only performed once, when the artefact, information, or service is provided. However, we might need an artefact that performs reliably for 30 years with only annual inspections and minor

maintenance, such as repainting. We might require information that will be valid for at least five years without requiring an update. We might need a service that can be provided in the same way 30 years later with no reduction in the quality of the service received by the end user. How can we be confident that an artefact that passes an acceptance test on delivery will still pass when tested 20 or 30 years later?

One option is to incorporate what is known as *accelerated aging* into the test specification. An artefact intended for outdoor application in a coastal environment could be subjected to alternating high and low temperatures, high and low humidity, intense ultraviolet radiation, salt spray, and vibration, perhaps simultaneously. Degradation that might take a decade or more in the actual location might be reproduced with two weeks of testing by using accelerated aging. Obviously, the method used to accelerate degradation depends on the application. Electronic circuits might be aged with a combination of heat, humidity, dust, electromagnetic radiation, and vibration. However, building a test facility to provide the required ageing in a reproducible manner is not cheap. Civil, hydraulics, mining, and aeronautical engineers who work with large structures often have to rely on scale model testing to keep the cost reasonable.

This kind of testing, therefore, has limitations. Another kind of specification, method specification, provides an alternative approach that can help overcome this weakness.

Method specification

The second kind of specification is known as *method specification*: we describe the method, process, or procedure by which an objective is to be achieved. A method specification will often include detailed specification of the tools, materials, and monitoring of the production process.

A method specification gives us confidence that an artefact will perform as intended by ensuring that it is constructed in a known manner by suitably trained and experienced people, it is made from materials of known quality, and it is inspected during the manufacturing processes using standardised techniques.

Sometimes, a specification requires a combination of these two approaches. For example, a method specification may require certain intermediate acceptance tests to be performed during construction, manufacturing, or assembly processes.

One of the most difficult aspects of writing a specification is to consider how you can be confident that the product or process will perform as expected throughout its service lifespan. Acceptance tests and inspections can be performed only once, when the product or service is initially delivered, before it commences operations.

There is a common weakness in acceptance testing procedures defined in specification documents that requires careful thinking and writing to overcome. Along with specifying methods and tests, the specification must also define what is to happen if any of the inspections or tests reveal non-compliance. For example, the hydraulic controls unit shown in Figure 9.5 was subjected to a long sequence of acceptance tests before leaving the manufacturer's factory. When the tests were performed on the first units that came off the production line, some of the tests revealed failure conditions, often associated with the bundles of black cables visible at the top of the photograph. The

test engineers requested that the cables be replaced before repeating the test that had produced a failed result. However, they did not return to the start of the test sequence: they simply repeated the failed test. If the unit passed, then they continued with the remaining parts of the prescribed test sequence. Many of these units subsequently failed within a few weeks of being installed at the end user's facilities, even though they were designed to have a service life of at least 30 years. The procurement engineers working for the project management organisation forgot to include a requirement to repeat all the tests after any repairs to rectify failures detected during the factory acceptance testing. The cause of the failures was unreliable cable connections. When the cables were replaced, new faults were introduced that were not detectable unless the entire testing sequence was started from the beginning.

Here are a few questions to elicit some further aspects of specifications:

- What is the predicted technical performance, safety, and environmental impact of the artefact to be created?

. . . or . . . for information or services to be provided:

- What are the technical attributes, safety, and environmental impact of the service or information to be delivered as a result of the project (including those as a result of its application by end users)?
- How can these be described in a meaningful way for the client?
- How can these be described in a meaningful way for the people who will create them?

For example, a client may want electrical cabling to be removed and replaced in order to upgrade the power handling capacity of the cabling.

To do this effectively, it will be necessary to specify many details that the client is not particularly interested in. Even if the client has not specified a particular standard for the cabling, the project team will almost certainly specify a national (or international) standard. There are many minor technical details that can be very time-consuming and costly to specify in detail. For example, we could specify every joint, bend, and junction in the conduit (tubing) that will enclose the cables after they have been installed. Alternatively, we could elect to specify AS3000, the Australian wiring standard. A contractor responsible for installing the cabling will then be able to work out all the details required for the conduit, because this particular standard provides all the necessary guidance, and you will be selecting a contractor who is already familiar with it.

What needs to be specified for this project?

This may require even more discovery performances in the form of discussions with the client or project sponsor, as well as other stakeholders, such as the engineers or contractors who will participate in the project. It is possible for a project to be over-specified, which will entail unnecessary, extra costs.

Which codes and standards will be invoked to guide the performance of technical work during the project?

While the requirement to follow particular codes and standards would normally be stated in a method specification, most test specifications also call for compliance with

similar codes and standards. These can be a combination of local, national, company, industry, and international standards.

Testing and inspections

Who will be responsible for performing any tests or inspections?

Tests and inspections will confirm that the project objectives have been met and the deliverables comply with the client's expectations. However, the supplier has an interest in passing the specified tests and inspections, so if these tests and inspections are performed by staff from the supplying organisation, there will be a tendency for them to portray the results in the most positive way possible. In the same way, engineers working for the procurement or project management organisation want to avoid, at all costs, a situation in which they accept equipment or services only to find that later, there are defects that they overlooked. Therefore, they may tend to interpret any test or inspection result in the most negative way possible, adding to the costs sustained by the supplier and possibly leading to contract disputes. It is also possible (and this has happened regularly) that the engineers working for the procurement or project management organisation have received inducements from the supplier organisation to adopt a lenient or generous opinion when interpreting tests and inspection results.

For these reasons, it is not unusual for acceptance tests and inspections to be performed by a neutral third party, one without an interest in portraying the results in any particular way.

- Is the person performing the acceptance tests or inspections a genuinely independent third party, someone who is employed neither by the client nor by the project team? In other words, is this a person who cannot readily be influenced by either the client or the project team?
- Has the involvement of this person been agreed on by the project team and the client, such that both will accept that person's judgement and evaluation?
- How can we describe the level of quality that the client expects in a way that is meaningful to the client?
- How do we describe this level of quality in a way that is meaningful for the people who will be performing the work?
- How can we measure or evaluate quality during the project, as well as at the conclusion?

Intellectual property (IP)

- Discuss any new intellectual property that may be or will be generated as a result of the project.
- To whom will this new intellectual property belong when the project is completed?
- How will any new IP that is generated be protected, and who will pay for this protection?

Keeping the client informed

What does the client expect to be told during the project?

- How are we going to keep the client informed about progress related to the project objectives?
- What happens if there are unexpected setbacks or unexpected opportunities?
- How and when do we inform the client about these?

There is always the possibility that a project may not succeed.

- How do we describe the client's expectations about the risk that the project might not succeed?

What documentation does the client expect to receive?

- When does the client expect each document to be provided?

Checking and reviewing the scope

Once you have answers to all these questions, you will have nearly completed your scoping document.

Provide your document to the client and ask the client to review the document carefully and provide feedback and suggested changes. This is yet another discovery performance: expect the subsequent discussion to provide plenty of learning opportunities.

Do not expect the client to send you detailed comments and suggestions for changes. This might happen, but often, it will not occur. Instead, be prepared to meet face-to-face with the client's representative and go through the scoping document line-by-line and sentence-by-sentence to ensure approval that the client is comfortable with everything written in it.

Write a list of all the changes that you have agreed upon with the client. Incorporate the changes and send the revised document to the client with a further request to check it carefully.

At this stage in many projects, there may be a renegotiation on the budget for the project, and also on the acceptance criteria: how you and the client will know that the project objectives have been achieved.

Close alignment usually requires extensive discussions to clarify differences of interpretation and to gain insight into the other participants' understanding of the objectives. A well-written scoping document usually only emerges after these types of extensive discussions.[11]

Stakeholders

Write a list of all the individuals, organisations, or other groups of people that are:

- actively contributing to the project,
- whose interests may be affected as a result of project execution or completion, or

- who may also exert influence over the project objectives and outcomes, as well as the ability of the team to achieve them, should they choose to do so.

Each of these people or groups of people is considered a *stakeholder*.

Examples of stakeholders:

- *Yourself – project manager or engineer in charge*
- *Client or project sponsor*
- *Project team members*
- *Competing engineering firms*
- *Contractors and unions*
- *Local community*
- *Media*
- *Regulators (in government departments)*
- *Suppliers: companies supplying materials, components, and hiring equipment*
- *Local government authority*
- *Social and environmental interest groups, NGOs (non-governmental organisations)*
- *Customer/user – note the possible multiple layers of customers, for example, a pharmaceutical product is prescribed by doctors and taken by patients who may be compensated by insurers. The customer may be the person paying for the project and the users may be separate people who use one or more products.*

Which of these 'stakeholders' is most affected by, or could exert the greatest influence over, the success of the project? (Key stakeholders) Identify the interests, values, requirements, and expectations of each stakeholder

In Chapter 8, we discussed values and interests in the context of motivation for teaching performances. Use the same concepts and examples here. Expectations are outcomes that are anticipated by a stakeholder, but which may not have been stated explicitly. For example, engineers assigned to an advanced technology project might expect to be involved with technology development. However, if they instead find themselves performing routine administration work in which the advanced technology plays no part, and the engineers require no understanding of the technology to perform their work, then they will perceive that their expectations have not been met. As another example, most clients expect to be informed immediately about any developments that significantly affect the likely project outcomes. They expect not to encounter nasty surprises in project review meetings.

How can you identify the interests, values, requirements, and expectations of each stakeholder? Express these in the language used by the particular stakeholder

Answering these questions requires discussions with each of the stakeholders. These discussions are further examples of discovery learning performances. It is crucial that you prepare accurate notes so that the statements of values and interests are constructed in the specific language of each stakeholder. It can be handy to include actual quotations so that other stakeholders can better appreciate the context of these words.

Examples of interests, values, requirements, and expectations:

- *Companies supplying materials, components, and hiring equipment*
  - *Interests: attributes of project outcomes that provide measurable benefits or penalties, e.g. i) business cash flow, ii) profit, iii) opportunities for publicity . . . 'Our product is being used for project X', and iv) opportunity to learn about new industry techniques.*
  - *Values: attributes of the business that owners wish to emphasise or identify themselves with, e.g. i) profitable business, ii) good customer service experience, iii) long-term business relationships, iv) ethical practice, v) development of local community, and vi) social justice or help for less fortunate people.*
  - *Requirements: i) payment within 28 days or the application of interest on outstanding accounts and ii) customers demonstrate an ethical record of doing business in the past.*
  - *Expectations: i) reasonable notice of delivery deadlines, ii) customer applies, stores, and uses products within the guidelines provided by a supplier, and iii) customer ensures products are only used by staff with appropriate training, protective equipment, and who maintain appropriate safety precautions.*
- *Regulators (government departments)*
  - *Interests: i) orderly conduct of business operations complying with both the intent and details of regulations, providing maximum community benefits and minimum harm to community interests, ii) good relationships with operating companies, iii) avoiding unnecessary disputes on regulations and time-consuming investigations, and iv) opportunities to learn about new industries and processes.*
  - *Values: i) maximising overall community benefits, ii) supportive environment for businesses, iii) long-term business relationships, and iv) ethical practices.*
  - *Requirements: i) businesses apply for permits and approvals well before they commence operations that require such approvals, and ii) applications for approvals comply with the state requirements and procedures.*
  - *Expectations: i) businesses demonstrate a commitment to not only meet, but also exceed regulatory requirements, ii) businesses maintain adequate records of operations to demonstrate both practical compliance and the intent to comply with operating requirements, and iii) business are open and provide immediate access to information and their operations following reasonable notice and requests.*

Explain the 'value proposition', defined as how the project outcomes add value or provide measurable benefits to the client and other stakeholders, as well as any costs or penalties

This question is crucial because the responses from discussions with stakeholders provide the essential understanding necessary to handle risks and opportunities: unpredictable events that will influence the project outcome.

How can the risks and opportunities (time schedule, financial, regulatory, community, environmental, health and safety, political) be shared amongst stakeholders?

Uncertainty cannot be eliminated, but it can certainly be managed, as we shall see in the later section on risk management. We can plan the project in a way that reduces the likelihood and consequences of risks that can negatively affect stakeholder interests. At the same time, we can also plan in a way that is likely to maximise benefits and opportunities that arise from unpredictable events.

Few engineering firms, and certainly not individual engineers, have the necessary financial capacity to be able to pay for the consequences when a project overruns its budget or runs late, thereby incurring financial penalties. Therefore, it is critical to negotiate the sharing of these risks so that when unfortunate events happen, stakeholders who are accepting the financial and other consequences are well-prepared. Naturally, these same stakeholders would normally be the ones receiving the greatest benefits when the project succeeds.

Understanding the ability of the project sponsor or client, in particular, to accept financial risks makes a big difference in the way that the project is planned.

For example, consider a project to develop a highly profitable product that can easily be copied by competitors. The project sponsor hopes to capture the market quickly and sell enough to make a reasonable profit. However, if the project runs late, there is a much greater risk that competitors will enter the market first. Therefore, the project sponsor can probably tolerate a significant budget overrun, provided that the product is ready on time. Therefore, the most important priority in the project will be to maintain the planned project schedule.

Alternatively, consider a project to refurbish a shopping centre in a mature property market. The shopping centre will have a long life with predictably secure financial returns from shop owners who will pay rent on long-term lease agreements. However, being a relatively low-risk investment, the financial return will also be relatively low, so it will be very important to restrict any budget increases as much as possible. On the other hand, maintaining the project schedule would be significantly less important.

In the former case, the stakeholders will be more prepared to accept a budget increase, whereas in the latter case, the stakeholders are more likely to accept late completion in order to avoid a budget overrun.

The most important aspect here is to discuss the risks with the stakeholders who are most affected to make sure that you have their support and confidence.

Chapter 11 provides a much more extensive discussion on engineering investment decisions.

What are the best locations for stakeholder discussions?

Stakeholders are more likely to feel respected if discussions take place at or near their 'home territory'.

How do we best maintain a relationship and communicate effectively with each stakeholder?

- What is the most effective communication channel: face-to-face, telephone, e-mail, documents, news media, or through community groups?

What can you and the project team do to influence the perceptions of each of the stakeholders?

Creating a positive perception would mean that each stakeholder sees the project outcome as providing benefits, and therefore contributes to the success of the project.[12]

Are there stakeholders who cannot be adequately represented at discussions about the project?

This might be the case because they are too far away, cannot communicate effectively, or are not yet present, as in the case of future generations.

The environment can be included as a stakeholder without a voice, although there will often be vocal groups in the community who will claim to represent environmental interests.

Ethical issues also present themselves in this context. With the knowledge and expertise we have as engineers, we are often in a privileged position to be able to anticipate situations where the benefits of a project may be shared unfairly. Professional ethics demand that we safeguard the interests of people, and the environment, that will be affected by any project but cannot be adequately represented to present their point of view when project decisions are made.

What will be the effects of the project work on the people who are involved in the project?

The effects occur either because they are employed to work on the project or because they live nearby, meaning that they will be close enough to experience the effects of the project.

What kind of meetings or negotiations will be needed to align the project objectives with the interests and values of all the stakeholders?

This is the sustainability agenda; we will learn more about this in Chapter 12.

What is the best way for each of the stakeholders to have some means of influencing project decisions during the course of the project, should they feel that to be necessary?

What special knowledge could stakeholders contribute to help make the project be more successful?

The work breakdown structure

Do you have access to the work breakdown structure for a previous, similar project?

Never start building a work breakdown structure from scratch if you can adapt aspects of documents produced for earlier, similar projects. Learn from others who have done that type of work before.

Write a comprehensive list of all the physical assets to be created and the work activities that need to be performed

The reference point for the work breakdown structure is, most likely, a digital CAD model of the objects to be created or services to be provided.[13] This data provides a detailed list of components that enable the definition of all activities and procurement that have to be completed in order to achieve the project objectives, as defined in the project scoping document. Include activities to be performed by all contributors, not just the project team. Include the review of documents by the client and processing of approvals by regulators. Use the responses to the questions in the scoping document as a starting point.

Each activity should normally take at least one day to complete. Activities that last more than 20 days (four working weeks) should be subdivided into shorter activities. Otherwise, it can be difficult to monitor progress on the project. We will return to this issue later.

Specify the location for each activity and physical asset. Almost always, work conducted at two or more separate locations should be included as separate activities, with at least one activity in each of the respective locations.

When calculating costs, distinguish between the direct cost of physical assets and the indirect cost of activities needed to get them in place.

Devise a way to group activities to minimise complex interactions

Do your best to arrange these activity items so that the list is organised and the activities are arranged in groups that make sense in the context of the project.

For example, activities might be arranged in groups according to:

- obtaining approvals and permits,
- the technical expertise necessary to perform them,
- the contractor who will perform them,
- the location at which they will be performed, or
- the phase of the project: early planning, project definition and agreement, preliminary design, detailed design, procurement, manufacture, commissioning, operating, or decommissioning.

Normally, a good project manager will group the activities to minimise the complexity of interaction between the work that needs to be performed in each group. Thus, as far as possible, difficulties experienced in one part of the project defined by a group of work activities will have the least possible effect on other parts of the project.

When you are evaluating the project, identify every activity in the WBS that was added to the list after the initial planning stage. The fewer activities added later, the better the quality of the initial planning and the WBS.

How long will each activity take to complete, and what will it take?

For each activity, estimate the duration and what will be needed to complete it in terms of human effort (hours of work needed), financial expenditure, and material resources. Also, identify other preliminary activities that need to be completed (either fully or partially) before each activity can be started (activity constraints).

If these estimates are uncertain, estimate the least likely value and the greatest likely value as well.

Prepare a bill of materials listing every item needed for the project and how it is to be obtained, transported, and stored

- Obtain Material Safety Data Sheets (MSDS) for every material that might be considered hazardous in any circumstance.
- Create a management system to ensure that every person handling hazardous materials has received appropriate training and wears appropriate protective equipment when handling the materials.

Arrange procurement, suppliers, and contractors

For every item that needs to be purchased or obtained from another organisation, either purchased or leased, identify possible suppliers or equipment hire companies. With their help, determine delivery lead times: how far in advance do the necessary purchase or procurement contracts have to be completed in order for the items to be available when they are needed?

Similarly, for each contract for supplying services to the project, identify a reasonable number of possible contractors who will be able to supply the services at the time they are needed. For each contractor, estimate the cost of the services and any support requirements that have to be provided by the project organisation (such as office space, desks, parking, special tools and equipment, and training).

Define the process by which the project organisation will choose which suppliers will provide the procured items and which contractors will provide the services. Make sure that the project plan includes activities for the project team to define this process, as well as to perform the necessary selection work and any negotiations that might be required. Also, consider prequalification: suppliers need to demonstrate that they can achieve certain standards before even being considered for selection.

Who will be responsible for monitoring each activity?

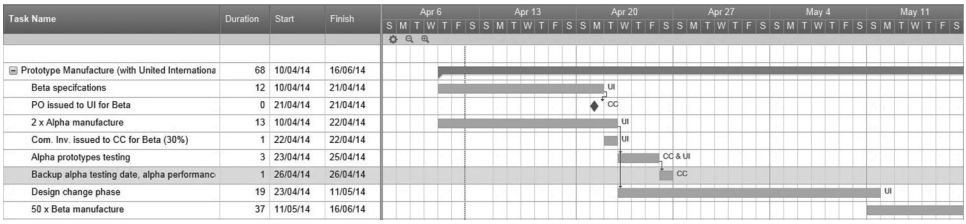
Wherever possible, assign a project team member who will be personally responsible for monitoring progress and making sure that every activity is successfully completed.

How would someone know that the activity is completed (are there deliverables or intermediate milestones)?

How long will the project take to complete and what is the likely cost?

Based only on the constraints of each activity, determine the most likely schedule for the activities required to complete the project. There are standard techniques for this, which are described in detail in project management texts and are available in software packages, such as Primavera, Microsoft Project, Libreplan, and many others.[14] Even Excel can be used with appropriate macro programming.

Using information about procurement items and services to be supplied by contractors, estimate the cash flow and payments required for each month (or a shorter period) of the project.



Task Name	Duration	Start	Finish
Prototype Manufacture (with United Internationa	68	10/04/14	16/06/14
Beta specifcations	12	10/04/14	21/04/14
PO issued to UI for Beta	0	21/04/14	21/04/14
2 x Alpha manufacture	13	10/04/14	22/04/14
Com. Inv. issued to CC for Beta (30%)	1	22/04/14	22/04/14
Alpha prototypes testing	3	23/04/14	25/04/14
Backup alpha testing date, alpha performanc	1	26/04/14	26/04/14
Design change phase	19	23/04/14	11/05/14
50 x Beta manufacture	37	11/05/14	16/06/14

Figure 10.2 Small part of a project Gantt chart.

There are standard ways to represent the activity schedule in diagrams. The most common (and in many ways, the most limited in usefulness) is the Gantt chart, which displays each activity as a horizontal bar with a time scale along the horizontal axis. Wikipedia has a wealth of reference material on these techniques.

Many project management texts focus considerable attention on calculating the likely time schedule, human effort requirements, and cash flow. However, this aspect of project management is almost routine and can be easily arranged with one of the many software packages that are now available. The difficult part is answering all the questions that we have discussed in this chapter: the responses lead to the data that you need for the software packages.

Project locations

Where is the project work going to be performed?

- How much space is required?
- Have members of the project team visited the site and surveyed the available space?
- What access is available to transport supplies and equipment to the location? How is this access likely to be affected by weather conditions, especially heavy rain or flooding?
- What is the maximum vehicle length that can access the site?
- What is the maximum weight of a vehicle that can access the site? What is the load capacity of the access route? What is the clearance, the maximum load size that can be accommodated by wheeled transport? Are there power lines that have to be lifted when large loads need to enter or leave the site?

  The Swedish government donated a 70-tonne demining machine to the Cambodian government in the 1990s. At the time, Cambodia had a major catastrophe with landmines left over from the conflict following the Pol Pot regime. Unfortunately, the Swedish authorities discovered that the bridge linking the port where the machine was unloaded with the rest of the country could only support a load of 50 tonnes. The machine had to be shipped back to Sweden until the bridge could be strengthened.

- Who owns the land, not only where the work is to be performed, but also the land immediately adjacent to the site?

- Is the land, or land in the immediate area (that may need to be accessed for transport to and from the location), subject to a contested claim or perhaps is it of special significance to indigenous or tribal people?
- Are there any significant resource constraints on the project, for example, limitations on the amount of water or land available for the project?
- If some activities are located in a foreign country, how will logistics, customs clearance, import permits, and dangerous goods be handled?
- How long will it take to obtain each of the approvals and permits for the project?

What kind of approvals will be needed for different kinds of operations at the site, as well as for transport movements to and from the site?

- What restrictions are in place that could affect operations at the site, including transport to and from the site?
- What are the weather conditions at different times of the year at this location?
- Where will the water required for the project (and for ongoing use) come from? What is the quality of the water? Is a separate drinking water (potable water) supply needed?

What is the sustainability of water and energy supplies in the area?

- How will used water and rainwater be disposed of? Will any of the water be contaminated, thereby requiring special holding ponds, treatment, or special disposal methods?
- How will sewage be handled at the site? How many people will be living on-site and do the existing sewage disposal facilities have enough capacity?
- If the location is subject to a wet season or occasional downpours, how will unusually heavy rain be handled? Is there a possibility of rain entering the site because of flooding, leading to the site becoming contaminated and the water needing to be treated or prevented from leaving the site? This is particularly relevant for mining operations in the northern part of Australia.
- What is the status of electric power supplies at the site? How much power will be needed for lighting, heating, cooling, site operations, and construction? Remember to include the power required for offices and living accommodation at the site.
- If additional power is required at the site, how will it be supplied? Is solar power an option? Can the site's power consumption be varied to suit a renewable source, such as wind or solar power? If a diesel backup generator is needed, how will the fuel be transported and stored? How will the fuel storage be protected to ensure that there is no contamination from leaking fuel?
- What kind of security fencing and access control measures are needed at the site?
- How will you communicate with people at the site? Does the existing communication infrastructure have sufficient bandwidth and how would you know? Will you need additional satellite communication capacity? Can an optical fibre be laid at the site, and if so, from where and at what cost? Are there any restrictions on electromagnetic radiation at the site (especially if it is in an area where there are radio telescopes or other sensitive instrumentation)?
- How will rubbish be collected and removed from the site?

- What kind of accommodation will be needed for people at the site? Will they need to work shifts? Will the workers fly in and fly out? Where is the nearest airfield to enable that to happen? What will be the transit time for workers travelling to and from the site? What kind of advance reservations will be needed to make sure that the necessary people can travel to and from the site when needed, possibly on short notice?
- How will work at the site be monitored and inspected when required? What will be the cost of monitoring the work and performing the necessary inspections?
- What kinds of excavation or drilling are required at the site?
- List all the approvals and permissions needed for operations at the site.
- What kinds of safety precautions will be needed at the site?
- Who will have access to the site and under what conditions? Will visitors to the site require special safety inductions, and will they need to be escorted?

How will medical services be provided at the site?

- Where is the closest hospital that can handle serious medical issues?
- How will people be transported to the hospital in case serious medical treatment is needed?
- What kind of training will be needed for staff at the site to make sure that adequate first aid can be provided?
- What kinds of health issues exist at the site for people or animals? What kinds of disease could people or animals be exposed to? What kinds of immunisation will be needed?
- What storage facilities will be required at the site?
- How will security at the site be maintained, both for materials and people?
- What kinds of recreation facilities are available close to the site?

Practice concept 65: Risk assessment and management: working with uncertainty

Identify risks and opportunities

As explained at the start, engineering work is intrinsically unpredictable to a certain extent because so much of it depends on interpretations and subsequent actions and performances by individual people. Each person involved in engineering projects may be involved in up to 60 or more simultaneous, overlapping activities, many of them not even on the same project. The attention needed for each of these activities can interfere with the work on all the others. Many, if not most, people in engineering organisations have a great deal of autonomy in deciding what they do each day and when they choose to start work on any particular activity for a given project. People can easily forget activities to which they are supposed to contribute, partly because they have so many disparate responsibilities.

There are other sources of uncertainty that can influence any project. Broadly, we can classify these as ‘opportunities’, which are unpredictable events that bring potentially beneficial consequences for a project, and ‘risks’ or ‘hazards’, which generally connote negative or undesirable consequences for a project.

Technically, 'risk' can be associated with a potentially advantageous event just as much as a potentially undesirable one. However, most people instinctively associate 'risk' with undesirable consequences. Therefore, it is probably more appropriate to use the term 'opportunities' to denote unpredictable events with potentially beneficial consequences.

Particularly in an engineering context, it is essential to create a broad understanding of the risks involved in any venture. It is particularly important for the engineering contractor to ensure that the client understands the risks and that appropriate allowances are made for them. This provides a way to share the risk, rather than expecting either side to carry the full liability.

Risk management has been executed successfully if there are no surprises along the way.

What are the unpredictable events that could affect project outcomes?

For each group of activities in the WBS, we need to identify intrinsically unpredictable events with consequences that could influence the outcomes, the time, or the resources needed to complete the activity group.[15]

Each event with potentially undesirable consequences can be listed as a risk. Each event with potentially desirable consequences can be listed as an opportunity.

Try to avoid grouping distinctly different events with different consequences or impacts on the outcomes. Note that this is not easy to do in practice.

Who can identify risks?

Very often, as engineers, we are not the best people to identify risks and opportunities. Perhaps it is because we tend to think logically and are often better at convergent, linear thinking rather than lateral thinking. Other people, even people without any technical background, can be better at predicting the unpredictable. For example, local community members can often be better at predicting events with the potential to harm the environment. The best strategy is to seek a diverse range of people to participate in risk identification.

Until around the year 2000, expert risk management consultancies were called upon for their specialist knowledge to identify the key unpredictable events that could affect project outcomes. However, research and public experience suggested that expert risk assessment was often wrong. For example, consider the UK-France Channel Tunnel opened in 1995. From the earliest stages of design, the possibility of a fire in the tunnel was taken very seriously. Designers went to enormous trouble to reduce the risk of fire and experts were happy that the risk had been reduced almost to zero by the time the tunnel was opened. However, within 12 months of opening, a serious fire broke out in the tunnel, causing several fatalities and closing the tunnel for several weeks. The subsequent loss of business was catastrophic for the tunnel consortium; the British and French governments had to rescue the project.[16]

There is increasing research evidence that organisations are less capable of assessing risks to their operations than we might expect. They are insufficiently able to learn from their own experience and find learning from the experience of others even more difficult. We do not understand the reasons for this, but the evidence is impressive.

Ultimately, the general public carry the cost of engineering failures. The Piper Alpha platform disaster cost Occidental (the platform operator) about 1/10 of the total loss sustained by the British Government and the wider public. Therefore, the recommended practice is to involve diverse stakeholders, particularly the local community, in wide public consultation on the assessment of risks. Researchers now recognise that non-experts can provide information that matches or exceeds the validity of that provided by internal experts. This is counter-intuitive, given the wide variation between subjective and objective perceptions of risk in ordinary people and experts. One would expect that objective risk analysis would always provide a better result than subjective assessment. Unfortunately, experience indicates that this is not the case; a careful combination of the two provides better results.

Needless to say, effective listening skills are essential for this collaborative discussion. Listening is all the more difficult for engineers who have devoted so much effort towards their favoured solution, especially if there is a risk that the solution might have to change to account for unforeseen risk factors.

Risk analysis

There are several techniques available in the literature.[17] Most project risk assessment uses qualitative descriptors, rather than quantitative probability analysis. Experience has shown that this approach works better for nearly all projects because there is seldom a sufficient amount of trustworthy data to perform a quantitative analysis.

Most organisations adopt a 'risk control' approach: risks (or opportunities) from unpredictable events are carefully and comprehensively defined, categorised, prioritised, and where possible, influenced by introducing control measures to improve likely outcomes. This approach is very useful in the context of engineering management, particularly for project management of a major contract. Before this approach was adopted, unpredictable factors were accounted for by adding a fixed contingency sum to the price of a contract, for example, 10% of the total costs. The percentage was decided on the basis of experience. Given that many engineering contracts are highly competitive and are priced with very small profit margins, mistakes in costing or judging the contingency allowance can make the difference between profit and loss. Replacing a fixed percentage contingency with a risk-based allowance can provide benefits for both the contractor and the client. Often, the client and contractor will agree to share different proportions of the risk for a given contract. For example, the client may agree to cover risks arising from unknowable factors, such as the nature of the ground deep under a construction site, whereas the contractor would normally agree to cover risks over which they can influence some control, such as late delivery of materials or equipment.

The engineering approach to risk management has been further refined in the field of reliability and maintenance engineering. In recent years, computers have enabled the use of quantitative methods for predicting the likelihood of failure. Complex computer models have been used on inventories of plant equipment listing, for example, every pipe valve and fitting in a process plant, together with the inventory of chemicals that they carry. The computer can readily identify hazardous combinations of materials and chemicals and predict inspection times to detect corrosion or wear that will be early enough to prevent a serious failure. Like other computer methods, this approach

relies on comprehensive data and accurate modelling. There is still considerable debate on the effectiveness of these methods: whether the savings in maintenance actually outweigh the costs of establishing the database, particularly in the case of an older plant for which construction data is not necessarily available. However, these methods are valuable for designing a complex plant to improve the overall reliability.

What is the likelihood of each of the unpredictable events?

We classify the probability of each event using a qualitative scale:

A: Highly likely within the project timescale, almost certain to occur
B: Moderate probability, probably will occur once or twice within the project timescale
C: Low probability, has been known to occur, but not on most projects
D: Remote possibility, has occurred on rare occasions

What are the consequences of each of the unpredictable events?

What are the classes of risk and opportunity consequences?

We classify the severity of the consequences (+ or −) into a qualitative scale:

1: Minor: relatively inconsequential
2: Medium: significant, but a small part of project value
3: Serious: significant proportion of project value
4: Major: several times greater than the project value
5: Catastrophic: very large compared with project value

Any unpredictable event can have different classes of consequences:

Reputation: the consequences can affect the reputation of the project team, as well as the clients and maybe other stakeholders as well. Reputation affects an organisation's ability to acquire financing for projects.
Schedule: the consequences will affect the time needed to complete the project.
Health and safety: the consequences can cause harmful effects to people working on the project, as well as end users, outsiders, bystanders, spectators, and community members. The consequences can be immediate or long-term; the latter is particularly relevant in the case of nuclear radiation or the release of contaminants into the environment.
Social, regulatory, and governance: the consequences can affect perceptions by members of the community that can consequently influence the actions of governments and agencies, such as regulators.
Financial: the consequences can affect the cost of the project or the level of benefits obtained from the project. The consequences can impact stakeholders unequally: some may be affected much more than others.

Tables 10.1 and 10.2 show examples of the criteria used to evaluate the consequences of undesirable risks for different companies involved in different industries.

Table 10.1 Sample risk consequences table for information systems in a health facility.

	Health	*Operations*	*Community reputation*	*Financial impact*	*Compliance*
Catastrophic (5)	Data error causing incorrect medication dose, patient has permanent health consequences	Incorrect code causes a system failure or loss of data affecting patient records	National current affairs investigation by newspaper or TV network	Corrective action causes operating loss for the year, permanent effect on financial return	Incident reported, restrictions on operating licence, temporary or permanent closure
Major (4)	Data error causing incorrect medication dose, health consequences, but patient recovers	Incorrect code causes system failure for two hours or less	Adverse newspaper article in national newspaper or on the front page of the local newspaper or TV news bulletin	Corrective action costs less than $1 million, ongoing financial impact significantly affects profit	Incident reported, health regulator conducts investigation, no ongoing effect on operating licence
Serious (3)	Data error causing incorrect medication dose, no health consequences	Incorrect code prevents system from being operated for 30 minutes or less	Single adverse article in local newspaper, not on front page, or multiple letters of complaint on a similar issue	Corrective action costs less than $100,000, ongoing impact less than 0.1% turnover	Incident reported with consequent follow-up by health regulator, no adverse consequences
Medium (2)	Systematic data error that medical staff can recognise and correct	Incorrect algorithm causes systematic errors that do not impact patient health	Patient or relative writes formal letter of complaint with a copy to government organisation	Corrective action costs less than $10,000, no ongoing financial impact	Incident needs to be reported to regulator, no follow-up required
Minor (1)	Isolated data error that medical staff can recognise and correct	Incorrect data causes incorrect billing for one or two patients	Single telephone complaint from patient or relative	Corrective action costs less than $1000, no ongoing financial impact	Incident does not need to be reported

Table 10.2 Sample risk consequences table for a chemical handling facility (financial not included).

	Health	*Safety*	*Environment*	*Community reputation*	*Compliance*
Catastrophic (5)	Debilitating lung condition to multiple people	Multiple fatalities or major third-degree burns to multiple workers	Contamination of local waterways of environmental significance	National current affairs TV program about chemical handling issues	Prosecution with the threat of losing operating licence
Major (4)	Permanent lung damage impacting standard of life	Totality or third-degree facial burns	Groundwater contamination	Repeated letters of complaint in regional newspapers affecting credibility	Licence breach with prosecution from government agency, significant fine, operating restrictions
Serious (3)	Bronchitis or breathing difficulties with hospitalisation	Chemical burns with hospitalisation resulting in lost time injury	Release into catchment pond	Group of neighbours request a meeting to discuss spill and response	Improvement notice received from environment agency
Medium (2)	Nausea and vomiting with local treatment	Minor chemical burns requiring medical treatment	Contained chemical spill immediately recovered	Neighbour writes formal letter complaining about spill	Incorrect incident reporting leads to late response
Minor (1)	Inhalation causing coughing	Skin and eye irritation	Chemical spill does not require specific recovery actions	Single phone call by neighbour to complain about chemical spill	Spill contained within licenced area

A single event can result in consequences with different ratings in the different classes. For example, failure to perform maintenance on a pump might lead to an unscheduled plant shutdown necessitated by a hazardous chemical leak arising from the failure of the pump seals.

The amount of leakage may be very small, perhaps detectable only by monitoring instruments within the plant itself.

The safety consequences are either none at all or minor, with no threats to the safety or health of any people.

The reputation consequences are negligible because the incident does not get reported to anyone outside the plant.

The social, regulatory, and governance consequences are also negligible because the incident does not get reported to anyone outside the plant, with the possible exception of environmental regulators, to whom the leakage may have to be reported.

However, an unscheduled shutdown of the entire plant just to replace leaking seals on a pump may have major financial consequences, particularly if the product from the plant has to be delivered on time to an important customer. Thus, the economic consequences of the seal breakdown are significant.

In evaluating the risk, therefore, we need to consider the greatest level of consequence in the different classes in order to decide the relative priority of dealing with this risk.

Given the likelihood and consequences, what is the risk (or opportunity)?

The normal approach is to use a lookup table to classify the risks according to the probability and consequence ratings for each unpredictable event.

L: Low
M: Medium
H: High
E: Extreme

Table 10.3 provides an example of such a look-up table, usually known as a 'risk matrix'. A project team will choose an appropriate risk matrix for their analysis. Most organisations adopt a standard risk matrix.

Table 10.3 Example of a risk matrix.

Consequence/ Likelihood	*1: Minor*	*2: Small*	*3: Medium*	*4: Large*	*5: Extreme*
A: Highly likely	L	M	H	E	E
B: Moderate	L	L	M	H	E
C: Unlikely	L	L	L	M	H
D: Remote possibility	L	L	L	L	M

What are the most important risks and opportunities to focus on?

The lookup table provides a simple measuring tool for the importance of each risk or opportunity. Usually, a project team will evaluate between 100 and 300 unpredictable events, but often far more on a larger project. The risk register, a spreadsheet containing details of all the unpredictable events, including their likelihood and consequences, is a critically important document for the project.

The simple classification of risks into four categories allows the project team to focus on the most important ones first. Usually, low and medium risks are disregarded or are handled with a single contingency fund.

How can we treat the important risks?

Finally, the risk treatment options are examined for every risk in decreasing order of severity. In any real project, the number of risks becomes very large, so this is a tedious process requiring attention to detail. Obviously, risks categorised as L, and possibly M, may not be worth considering for treatment depending on the number of high or extreme risks that need to be considered. The options are listed below. Sometimes, a combination of options is chosen for a particular risk. The costs for each treatment need to be clearly specified.

Option 1: Do nothing

We accept the risk, although we also possibly make a financial provision for it. This is often called a 'contingency'. In a contract, there might be a 'contingency amount' set aside to cover unexpected events, such as delays caused by bad weather, delays in the arrival of materials outside the control of the supplier, etc. The contingency amount might be 5% or 10% of the total contract value. Often, an external person (arbitrator or commercial facilitator) will be called in to decide whether the contingency amount can be used to cover the consequences of an event if the stakeholders cannot agree. A contract agreement might also share the benefits of achieving a lower than expected completion cost between the contractors and the client.

Option 2: Reduce the probability of the risk

We can sometimes do this through implementing technical precautions, using different materials, providing safety barriers, and utilising instrumentation to provide early warning, as well as sensors to detect when people are likely to take unnecessary risks.

We can also reduce the possibility of unpredictable events by obtaining more data, carrying out more detailed surveys, and performing more analysis. However, this comes at a significant cost.

There are many other options for reducing the likelihood of an event that arises from human behaviour. Most of these provide both productivity improvements and risk reduction.

- Planning and organisation
- Designing simple, effective procedures

- Checklists to reduce the risk of forgetting items
- Adopting standards, following standard procedures, and complying with relevant codes
- Training and practising with staff in using procedures
- Supervision and frequent reminders to reduce 'risk-taking behaviour'
- Checking and review by other people
- Configuration management – accurate documentation
- Coordination and monitoring
- Teamwork, backup plans, and redundant equipment
- Regular testing and inspection, with frequency in proportion to the risk (risk-based inspection)
- More frequent and diverse data backups to reduce the probability of data loss
- Increased frequency of progress payments by client to reduce the chances that engineering work will be performed when the client cannot pay

Option 3: Reduce the consequences

We can use personal protective equipment (PPE), such as helmets and eye and ear protection.

We can include protection or safety measures in the design, or use energy-absorbing materials, etc.

Other examples of undesirable consequence reduction measures include:

- Creating a low dam wall (or bund) and sealing the ground around a storage tank to reduce the consequences of a spill from a tank: the fluid is retained in the area immediately around the tank and does not leak into the ground.
- Providing automatic fire extinguishers to reduce intensity or extinguish an accidental fire completely.
- Arranging for payment in advance before undertaking work of equivalent value: if the client cannot pay, then work will not be started.

There is a possibility that a particular manufacturing process will result in a significant proportion of defective parts.

One way to control this risk is to anticipate that, for example, up to 10% of the parts will be defective; therefore, order 10% more parts than are needed, so at least the required number are likely to be available.

Another complimentary way to control this risk is to provide for defective parts to be repaired once they have been identified by quality control inspection. The latter control method requires an alteration to the project activity sequence: additional activities need to be inserted.

Of course, in the case of opportunities, we want to increase the consequential benefits, such as a potentially valuable discovery or innovation:

- Ensure that innovations that prove to be effective are rapidly and effectively shared with other divisions of the company. This is almost always best achieved by the people who implemented the innovation successfully (not necessarily the inventors). Plan on transferring these people to other company divisions (temporarily)

to help others use the idea. Also, consider patents, registered designs, and other forms of intellectual property protection.

Option 4: Avoid the risk

We can avoid a risk by choosing a different strategy or solution to a problem (applies only to negative risks).

Option 5: Transfer (or share) the risk

We can pursue this option by taking out insurance (for negative risks), by agreeing with another stakeholder for that stakeholder to accept the risk, or by agreeing with other stakeholders to share risks and limit the extent to which each will cover the risk.

- Take out an insurance policy to cover the risk: this transfers the risk to an insurance company, but you have to pay a fee (called an insurance premium) to persuade them to do this. The fee will be determined by their assessment of the probability of the risk, plus their assessment of the cost of administration and the likely cost of investigating any claims you might make. This means that insurance is often an expensive option.
- Share the risks between several stakeholders. This is now almost a universal practice on larger engineering contracts and much of the negotiation is not so much on price as on the ways that risk will be shared. The two extreme cases are:
    - Cost-plus contract: the client pays the costs incurred by the contractor, plus a small agreed upon profit margin. The client carries all the risk, so the profit provided to the contractor will be small. It may be possible to negotiate a decreasing profit margin for work extending beyond an agreed completion date to ensure that the contractor has an incentive to finish the project on time.
    - Fixed-price contract: the client pays a fixed price, regardless of the outcome. The contractor makes a big profit if things turn out well, but may take a loss otherwise. This works well on routine engineering work where the costs are predictable and controllable.

Usually, the risks are shared according to an agreed formula between the ultimate client, the main contractor, and the sub-contractors. Clearly, the amount of risk being shared depends ultimately on the financial strength of each party, because it is not in anyone's interest for one of the parties to go bankrupt as a result of a project failure. There are many ways to structure projects to mitigate commercial risks; however, these techniques are beyond the scope of this introductory chapter.

It is all too tempting to transfer the risks to an organisation that is prepared to accept the risk, but actually may not have the capacity to handle it if anything does go wrong. In the end, the result of that situation is that the transfer is ineffective because the task of handling the risk will fall back on the party that transferred the risk in the first place. For example, if transferring a risk to a contractor results in the contractor going bankrupt, then the work will have to be completed by the project organisation, often at a much greater cost. This is a common project pitfall.

How do we need to modify the project plan to build in the risk control measures?

Once we have designed control measures for all of the major risks, we need to go back to the stakeholders, particularly the client, and negotiate changes to the project plans and possibly the project budget.

Here, it is important to understand the value proposition that we mentioned earlier. How does the client, in particular, as well as other stakeholders, benefit from the risk control measures that we have planned? While it is tempting to reduce everything to a financial measurement of risk, this is usually not possible. The best guide for these negotiations is the homework that we did earlier when we identified interests, values, requirements, and expectations for each of the stakeholders.

Practice concept 66: Approvals

Engineering has become critical in supporting the social structure of our civilisation, with a large influence on the lives of every human being. Engineering projects can cause significant, unintended consequences for the community. Because of past failures, engineers now find themselves under detailed scrutiny by government regulatory agencies, which mostly employ engineers. Today, engineers find themselves devoting considerable effort to writing applications seeking approval for their projects from government agencies.

Even a small project can require 20 or 30 separate approvals, both within the project organisation for the plan and the budget, and from government regulatory agencies. A large project in an industrialised country may require as many as 150 or even 200 separate approvals from several different local, regional, and national government agencies.

The kind of planning we have discussed in this chapter provides the groundwork for these applications. Most agencies use some form of 'due diligence' criteria in assessing applications. In other words, has the project team considered all the relevant issues at a level of detail that would be appropriate for a well-run engineering project? Has the team conducted sufficient consultations with stakeholders, particularly all the people in the community who will be affected by the project?

If the planning has been done well and comprehensively, if the stakeholders have been involved in setting the project scope and constraints, including identifying, managing, and sharing the risks, and if these discussions have been appropriately recorded and summarised in the form of documents, then all the supporting evidence for most approval applications will already exist.

Once again, the most appropriate strategy is to consider the regulators as stakeholders in the project. Rather than preparing an application for approval towards the end of the planning phase, a good project leader will involve the regulators as stakeholders in discussions from the start of the planning phase. Developing constructive and helpful personal relationships with individuals in the regulatory agencies early in the project can avoid tedious objections and what seems like petty obstructive behaviour when an application for approval is rejected.

This is another kind of collaborative performance that draws on the ideas presented in Chapters 7 and 8.

What does the regulator expect in an application for approval?

Like other stakeholders, regulatory agencies have their own interests, values, requirements, and expectations. By listening early on in the project, it is possible to understand how they see the project. This understanding will help you write an application for an approval or operating licence that takes their expectations into account, written in a language, an English sub-dialect, which is meaningful for the regulators.

A good place to start is to find a previously successful application that was approved with only minor, if any, amendments.

Final Investment Decision (FID) approval

The most critical approval is the one that authorises real work to commence on the project, beyond the initial project definition and planning that has been described up to this point in the chapter.

Experience shows that projects with detailed plans prepared are much more likely to achieve their objectives. The concept of '*front-end loading*', developed by Merrow, reflects this. Front-end loading represents the degree of effort devoted to planning and preliminary design for a project in order to maximise the chance of achieving the stated objectives.[18] It is reasonable to expect that 10% of the project budget will be spent *before* the final approval to proceed is given. If the amount is substantially less, and the project is not one for which detailed plans could be adapted with minor changes from an earlier project, then you can expect to encounter trouble ahead.

Once final approval is given to proceed, work on the project activities listed in the work breakdown structure can commence and the monitoring stage of project management also commences at that point.

MONITORING PROGRESS

Once the project work commences, project management shifts from planning to monitoring.

As explained before, the essence of project management work is coordinating collaborative work performed by many different people, possibly in many different locations. In order to do this, it is essential to monitor the progress of every activity, which is not as easy as it looks.

Practice concept 67: Monitoring is a cyclical process

Monitoring requires a cycle of activity. The cycle is repeated regularly, but the time interval depends on how the project is being managed and by whom.

Many describe the monitoring process as a simple, endless, repeating cycle: plan, do, check, act, plan … One starts with a *plan*, and then people follow (*do*) the plan, someone *checks* the results against expectations, one takes any necessary corrective *action*, and then adjusts the *plan*, if necessary, repeating the cycle until the project is completed.

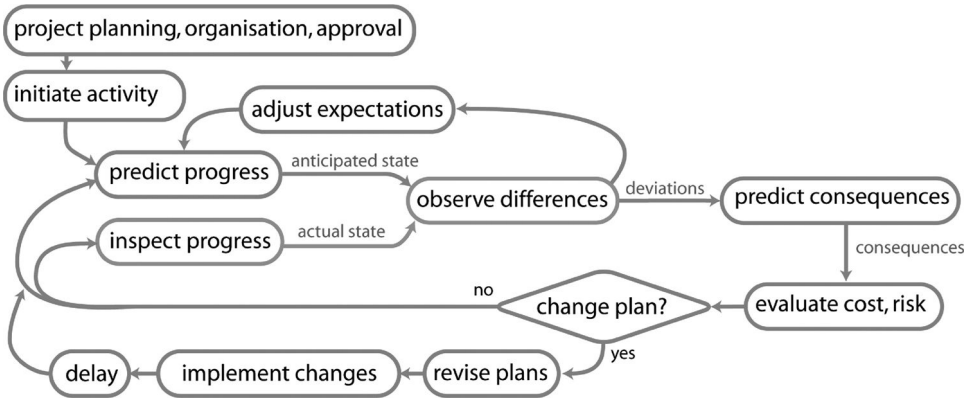


Figure 10.3 Project monitoring is a cyclical activity, identical in principle to the monitoring phase discussed in Chapter 9 on technical coordination.

Figure 10.3 illustrates the same idea: you will recognise this diagram as one from the previous chapter. In principle, there is no difference between the formal monitoring needed for managing a project and the informal monitoring needed in technical coordination. However, since a project involves many distinct activities that happen simultaneously, we need a different way of organising the monitoring.

Any decision on monitoring frequency will depend on the criticality of a given activity. Activities on the critical path, where any completion delay will delay the entire project, are obviously monitored more carefully than ones with several weeks of float time. An expert project manager will assess each activity and allow for the time needed to fix the consequences of foreseeable, but unpredictable, events that could affect schedules.

Monitoring occurs simultaneously with different levels of detail, scope, and frequency.

At the site where the activity is taking place, a foreman or supervisor will check on progress at least once a day, but usually two or three times. If there is a site engineer, they will review progress with the foreman or supervisor every day. Note that the site engineer is often younger and inexperienced: part of the responsibility of the foreman or site supervisor is to help the site engineer learn all about managing a site where there is practical engineering activity taking place. This can be a design office with several CAD workstations operated by drafting experts, a construction site, part of a factory, a software production house with programmers, a maintenance workshop, an electrical substation, or a power plant: there are countless different locales where engineers learn how to deliver real results.

A project engineer or project manager who is not located at the site will typically check on progress every week. They will visit the site and spend the first hour or two with the site supervisor or site engineer, working through every activity currently in progress and reviewing records of progress. Then, together, they will walk around the site looking at locations where there are hold-ups, unexpected issues, or mistakes that

need to be rectified, as well as simply conversing with sub-contractors, craftspeople, tradespeople, and technicians.

Perhaps you remember from the brief description about C. Y. Connor in Chapter 4, how he spent days at a time, up to 12 hours each day, inspecting work first-hand on the goldfields pipeline and storage dam at Mundaring.

Before each inspection visit, a project manager will predict the results from each currently running activity: what they expect to see. By comparing actual progress with expected progress, one can 'calibrate' one's expectations so that predictions become better and better over time.

Critical to all of this, of course, is establishing some way of assessing progress on each activity. Anything that results in tangible production is relatively easy to monitor. The difficulty with a lot of engineering work is that it is intellectual and cannot be directly observed. Intermediate results often give little indication of actual progress. Design work ultimately emerges in the form of drawings and documentation, but these appear relatively late in the process. In the early stages, the best one can expect to see is often just a series of sketches backed up by conversation with the designer. Here, one can begin to appreciate how important it is to think in advance about the questions posed earlier in relation to each activity.

How can we assess progress, quality, and other attributes of the work in question well before it is actually finished?

How can we tell if it is going to be completed on time, at an acceptable quality, on budget, and sufficiently long before it is actually needed, so corrective action can be taken, if required?

And finally,

How can we tell that it actually has been completed with sufficient quality, accuracy, and within the allocated budget?

One of the most difficult aspects of monitoring progress is a tendency for many engineers (and others) to engage in self-deception, particularly with engineering technical work that is highly intellectual and depends on abstract thinking, such as design and planning. William James foresaw this in his essay, 'Sentiment of Rationality'.[19] He distinguished between abstract rationality and concrete rationality. The former is more an emotional sense of satisfaction, a sense of inner peace, or the feeling that one has reached a resolution and that it all fits together rationally. The latter is 'reaching what is right', and requires that all the details be resolved in a way that others can comprehend sufficiently. Often, we only discover how much more there is to resolve when we attempt to convert abstract rationality into concrete rationality in the form of drawings and documents.

Many design engineers tend to underestimate the proportion of the design and planning work that has actually been completed. They create a reality in their minds in which everything has been resolved, except for the final stage needed to produce concrete reality: the drawings and documents. I have often experienced this myself. We will assure others that the work is essentially complete, maybe 80% done, without the slightest doubt in our minds. Only someone with an intimate knowledge and deep

experience can ask us questions to expose the gaps that we convince ourselves have been covered and resolved.

Often, a project manager does not have the depth of technical expertise in every aspect of the project to be able to ask these questions, which means that they go unanswered. Yet, as time drags on, we gradually realise that the 80% completion estimate was more like 30%. As weeks turn to months (and sometimes even longer), and completion times are pushed out further each week, the project gradually unravels in front of us.

The difficulty is that we were not lying when we said that the work was 80% done! We completely believed that at the time. We could have sworn an affidavit in front of a judge without the slightest twinge of doubt.

As a project manager, how can one detect these misleading situations? Particularly if one does not have the same technical expertise as the people one is supervising?

The first step, of course, is to ensure that there are appropriate supervisors with the technical knowledge to ask the questions that need to be asked.

Another technique is to carefully monitor the actual expenditure of time and resources on each activity. Most engineers are comfortable completing timesheets each week to account for their hours: they report the approximate number of hours spent on different aspects of one or more projects on which they are currently working. Even though engineers are notorious for misreporting their hours, carefully working out in advance what they are expected to report and then providing information that aligns with expectations, rather than necessarily with reality, this data can still reveal a lot of useful information about progress.

If the actual expenditure in money and hours, compared with the budget, reflects 30% progress, while the people performing the work are simultaneously reporting 70% progress, then there is a clear discrepancy. Very few engineering tasks can be completed that far under budget. That is a major warning sign.

Another way is to carefully monitor scope creep: an issue that we mentioned earlier. Scope creep appears in the form of a few additional documents that appear on the list of deliverables, or even additional activities that were somehow overlooked at the planning stage. Here, a good project manager learns to distinguish between activities that were accidentally forgotten at the planning stage and activities that have been introduced into the plan because of technical or other complications that were not fully anticipated or appreciated earlier in the project. The former can be covered by an overall contingency: an established allowance because some aspects of the project will always be forgotten. The latter, however, can be warning signs of a project that is getting out of control. It takes some time for the information about additional documents or activities to be included in project plans. By the time they are noticed, there could be even more requests for additions, more scope creep that is already in the pipeline. What initially seems like a few overlooked details becomes an avalanche: it is revealed that whole areas of technical complication were overlooked because of insufficient time to explore the details or insufficient experience with the technical aspects of the project.

Also, as I have explained, those closest to the technical work itself can often be the least able to appreciate the reality of the situation.

One progress monitoring technique that has emerged in the last decade is known as 'earned value' monitoring. Each activity forms part of the completion value of the

project in proportion to its total cost. A certain proportion of the activity value is said to have been 'earned' when the activity passes certain milestones, such as 50%, 80%, and 100% completion. For example, the project team may agree that 40% of the activity value has been earned when it is assessed as 50% complete. By carefully comparing actual spending on the project with the total 'earned value' from all the activities that have passed these milestones, a project manager can foresee future delays and budget 'blow-out'. If the earned value is significantly below the total spending, then a delay or cost overrun is likely.

What is the current predicted project completion plan (dates, expenditure)?

Once each round of inspections is completed, the project manager needs to update all the future predictions: when each activity is likely to be performed and when the project is going to be completed, as well as the latest estimates of expenditure. It is also necessary to review and update the project risk assessment. Each of these tasks can be tedious, but they are essential pieces of documentation to monitor progress. This is where computer tools can be really useful on larger projects, by updating these predictions automatically.

If there are delays (or opportunities), is it worth changing the plan?

At this stage, the project manager can evaluate the consequences if activities are running late or over budget. If the situation is likely to seriously compromise the interests of influential stakeholders, especially if there are changes in the likelihood or consequences of significant risk factors, then it may be necessary to modify the project plans.

Changing the plans inevitably introduces even more of a delay. First, there is the time required for everyone who is affected by the change to provide feedback and comments on the implications before any final decision is made. Particularly when equipment is hired and transported to a remote site, or when work is held up so that contracting teams have to stay longer at a remote site, the consequential costs can be very significant.

Once the plan has been changed, it takes time for everyone to become familiar with the revised plans and adapt their activity schedules accordingly.

So, as we can see from this brief discussion, the monitoring phase of a project, with its repeated prediction, inspection, evaluation, and review steps, is time-consuming. A project manager's job is not easy. Also, it can be difficult for others to appreciate the amount of time and effort involved. To those people performing the technical work itself, project managers often seem like expensive sources of overhead. Spending money on a project manager is often seen as a waste: 'It would be better to spend the money with us; that way we could achieve much more!'

Practice concept 68: Contract variations

Significant changes in the project plan that require new work, or rework that was not accounted for in the original estimates, will often lead to negotiations on how the additional cost is to be covered. The client will often argue that the work should have been foreseen by the engineers and covered in the project contingency, which is

the item in the project budget that covers uncertainties. These issues are referred to as *variations*.

In most projects, any addition to the scope required by the client becomes a 'variation'; it is not uncommon for the costs of variations to be estimated at a much higher level than the original work. Mostly, the additional costs are fully justifiable. The original work will have been carefully planned so that the specialised tasks requiring skilled labour have been grouped together. Bringing people with special skills and equipment back onto the site, perhaps for a job taking just a few minutes, can impose a large cost, particularly for a remote site. The situation is even worse if other people are in the way and the skilled people have to wait around doing nothing, simply because they can't access their equipment to perform the work.

In some industries, project delivery firms will negotiate a low price, perhaps even below the actual cost of the project, knowing that the client will request variations on which they can subsequently charge a much higher price so they can make their profit.

Variations almost always result in unpleasant cost increases for the client. By spending more time with the client during the planning phase, you can ensure that all the requirements are included in the project scope documents and specifications before the project commences. During the project, the client needs to be repeatedly reminded not to make any further changes – clients can be their own worst enemies in this respect when they make late changes to the scope of the project. If the client resists this temptation, then there is less chance that variations will be needed and a greater chance that the client will be happy with the outcome. Happy clients lead to repeat business and an enhanced reputation that brings work from new clients.

One common trap for young engineers is to try and be 'nice' with a client by accepting what seem to be minor scope changes without first checking the implications with more experienced engineers. Without painful experiences in the past to learn from, it is easy for a young engineer to overlook some of the consequences that can result in painful additional expenses that the client may not pay for.

COMPLETING THE PROJECT

Practice concept 69: Project completion is also a cyclical process

A project is completed when all the agreed upon objectives have been achieved and accepted by the client. Completion is relatively straightforward if the project definition phase included all the necessary details on how the completion of every task was to be certified with appropriate acceptance tests and inspections. Inspection and testing activities would normally be included as an activity in the work breakdown structure. Therefore, completion of the project corresponds with the completion of all the activities in the work breakdown structure.

Figure 10.4 illustrates the kind of repetitive process needed for the completion of each activity.

Once the person responsible for the activity tasks announces that they have been completed, the project manager would normally review the scope in order to anticipate what should be evident at task completion (the anticipated state). The project manager

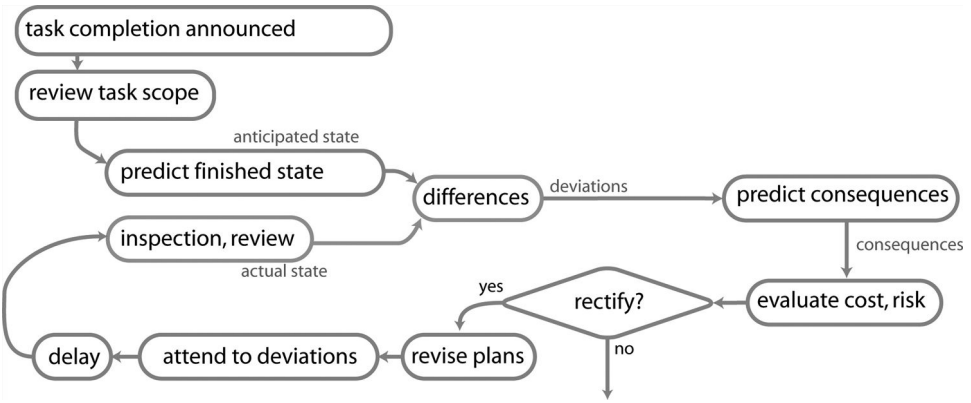


Figure 10.4 Activity task completion.

will then perform an inspection to determine the actual state, will have tests conducted, or will inspect independent tests and inspection reports to verify task completion. Depending on the contract arrangement, the client may also be responsible for verifying task completion.

Any differences between the anticipated state and the actual state are noted as deviations. All of the deviations noted for a particular task will be entered into a *punch list*, which serves as a record of outstanding deviations. The project manager, collaborating with the client for important issues, will then review the punch list and the consequences of leaving deviations as they are. Depending on the consequences, items on the list may have to be rectified, in which case the project plan is updated to indicate that the activity tasks are not yet completed. Additional time has to be scheduled for the deviations to be rectified.

Once they are reported as having been rectified, some or all of the tests and inspections will be repeated before certifying that the activity has been completed.

In reality, most projects end up with substantial lists of items that need to be inspected or rectified. The benefits of detailed documentation at the start of the project now become apparent: without details, a project manager can easily end up in protracted negotiation with the client on how to complete each activity. Naturally, the project manager wants to keep the overall expenditure within the budget or as close to the budget as possible. The client, on the other hand, has a vested interest in minimising payments to the contractor (the project organisation).

Normally, the project manager is under intense pressure to deliver the agreed objectives within the original project budget. Unless project documentation is sufficiently detailed, this tension presents the project manager with the tempting option to remove non-essential items from the scope, or to simply ignore them, particularly if they will cause the project budget or completion timescale to be exceeded. It is not easy for a project manager to negotiate an extension to the budget or time schedule as the project draws to a close unless there are well-established reasons that were documented earlier in the project.

This is perhaps the most fundamental weakness of the conventional method for organising a project. Project managers are rewarded for completing projects on time and budget, which can conflict with the interests of the organisation that inherits the assets and other deliverables created by the project. The next chapter describes a case study that illustrates this problem. A large process plant was constructed at a remote site by a team that did not fully understand the technology with which they were working. As the client's engineering team became more aware of the operating requirements for the plant, engineers in the client's organisation insisted on many changes late in the construction. For example, they insisted that adequate access be provided for maintaining equipment, such as the pumps. While this might seem to be a normal and reasonable requirement, these details had not been included in the original project documentation. Eventually, the construction project team ignored further change requests from the client, effectively locking them out of the site until all the construction was completed. When the client's operating team eventually took over the plant, there was a long series of major and minor plant upgrades to address all the issues that had been raised by the operating team. These upgrades were still not completed by the time the plant closed a few years later. With the plant already constructed, it was much more expensive to carry out these upgrades, but the operating budget only allowed for limited engineering work to be performed.

In our research interviews, engineers reported on several occasions that project managers, who had apparently completed projects on time and budget, had subsequently been rewarded with responsibility for even larger projects. However, they only managed to do this by deleting major items that would otherwise have caused a delay or a budget overrun. In doing so, they had been able to exploit weaknesses in the initial documentation of the project: gaps and inconsistencies in the project scoping documents and specifications allowed the responsibility for these items to be transferred from the project team to the operating organisation that would inherit them after the project completion. For example, access platforms and lifting devices needed for maintaining pumps and other equipment can easily be removed from the construction phase of a process plant without affecting its ability to operate for the first year or so. In doing so, the project team passes responsibility for installing the access platforms to the operating organisation: the platforms will be needed eventually.

Several improvements to project contracting arrangements have been proposed to prevent such transfers from happening. For example, a client can provide incentives that reward a project delivery team if the operating cost expectations are met for the first twelve months but these provisions can be very difficult to negotiate and evaluate. Here's an example from the offshore petroleum industry. The interests of the client organisation, typically specialising in operations and selling petroleum products, are:[20]

- Maximised shareholder returns
- Contractual obligations to customers (maximising profit and minimising penalties/damages for non-performance of obligations)
- Plant/facility efficiency with minimal downtime
- Minimum OPEX (operating expenditure)
- Minimum CAPEX (capital expenditure)
- Safety

On the other hand, the interests of the project delivery team are:

- Maximised shareholder returns
- Contractual obligations to clients (maximising returns to shareholders and likelihood of repeat business)
- Lowest possible expenses to achieve objectives
- Completion in minimum time
- Minimal risk (reducing possibility of claims)
- Safety

Even though there are clear conflicts between the interests of the client organisation and the project delivery team, there are also common interests, including safety, completion in the minimum time, lowest possible expense, and the likelihood of repeat business in future. The last item, in particular, is much more likely if the client and the project delivery team reach a happy compromise. The best way to achieve this is to spend more time on the project definition phase – maximising front-end loading.

NOTES

1. Some non-engineers might contest this, citing the common stereotype that many engineers are far too predictably boring, without necessarily understanding what engineers do and how we think.
2. Human error in organisations has been extensively studied and engineers can learn from this literature. For a discussion of operator errors, see Kletz (1991). Dekker (2006) provides a more modern discussion from an organisational standpoint. Other informative references: Bea (2000), Bigley and Roberts (2001), Bloch (2009), Busby (2006), Hobbs (2000), Klein, Bigley, and Roberts (1995), Leveson and Turner (1993), Malone et al. (1996), Reason and Hobbs (2003), K. H. Roberts (1990), Stewart and Chase (1999), S. Tang and Trevelyan (2010), and Weick (1987).
3. Mehravari (2007) and other unpublished studies by the author.
4. An experienced engineer reviewing this chapter commented that, in his experience alone, this factor had accounted for numerous multimillion dollar claims against his engineering firm.
5. Consultation with all stakeholders, including the local community, is the recommended standard practice reflected in ISO31000 and national standards (e.g. Standards Australia, 2009, pp. 14–16).
6. There are differing views on the need for technical capabilities amongst project and organisational managers (e.g. Grant, Baumgardner, & Shan, 1997; Hysong, 2008).
7. Project management terminology in this chapter has been set in bold type to help you find explanations for the terms later.
8. Sule Nair (2011), Unpublished manuscript.
9. Definition based on the *Project Management Body of Knowledge* (Project Management Institute, 2004, p. 5), see also (Hartley, 2009).
10. (Whyte & Lobo, 2010).
11. (Merrow, 2011).
12. (Project Management Institute, 2004, p. 26).
13. Whyte and her colleagues have provided insightful descriptions of what this entails in construction projects. Their accounts detail aesthetic issues, project coordination around fully detailed digital models, and also the limitations of these models when physical prototyping

is necessary (Ewenstein & Whyte, 2007; Whyte, 2013; Whyte, Ewenstein, Hales, & Tidd, 2008; Whyte & Lobo, 2010).

14. Hartley (2009, Ch. 6) provides some further details.
15. Standards Australia (1999, 2009, 2013).
16. Waring & Glendon (2000).
17. E.g. Standards Australia (2013).
18. Merrow (2011).
19. James (1905).
20. David Roberts (2008), Personal communication.